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KANNADA DIGITAL INNOVATION HACKATHON 2026 · TRAVEL & TOURISM

Plan Karnataka your way.

Dari (ದಾರಿ) builds a day-by-day Karnataka itinerary from real places only, with a route on the map, an honest budget and Kannada phrases for every stop.

No sign-up · Free to use · Works in Kannada

PARTICIPANT INFO

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PROBLEM STATEMENT

The Problem

THE CHALLENGE | Heritage | Food | Nature | Beaches

"Develop an **AI-powered travel planner** that creates **personalized Karnataka itineraries** based on a traveller's **interests**, **available time**, **budget**, and **preferred type of experience**."



01

Who is affected

First-time and domestic travellers, and the many visitors who speak no Kannada, arriving in a state with six very different tourism zones.

02

What goes wrong

Invented monuments, guessed ticket prices, and days that pair sights four hours apart. One wrong fact costs a whole day on the road.

03

Why it matters

Karnataka holds four UNESCO-listed sites and a 320 km coast, yet the planning layer keeps sending everyone to the same five places.

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EXISTING SYSTEM

How It Is Solved Today



Today's tool	How it is used	Where it breaks
Generic AI chat	Ask ChatGPT or Gemini for an itinerary	Invents places, fees and drive times; no map, no budget maths
Blogs and listicles	Hand-written "Top 10" posts	Static and generic; prices and timings go stale
OTA packages	Book a pre-built tour whole	Fixed and commission-led; thin beyond the headline sights
Google Maps	Route between places you choose	You must already know where you want to go
Tourism portals	Official place and season information	Reference material, never a day-by-day plan

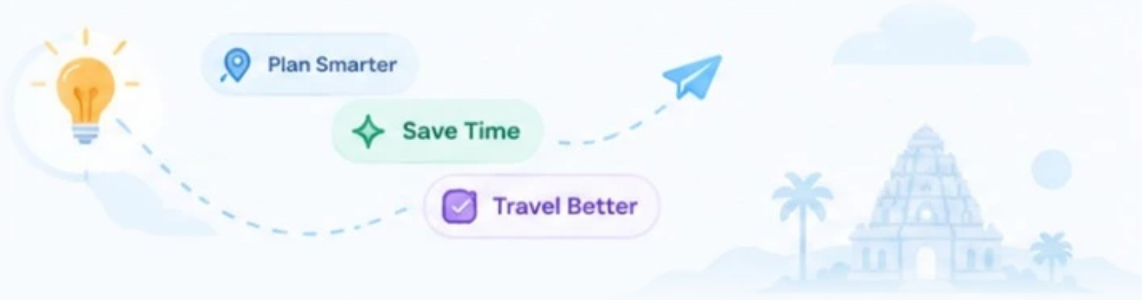
THE COMMON GAP

None of them speak Kannada, none check that a day is geographically possible, and every AI-backed one simply fails when the model is busy or rate-limited.

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HOW YOUR SOLUTION IS DIFFERENT

What Dari Does Differently



Feature	Generic AI planner	Dari
Places	Free text — can be invented	Only IDs from a 39-place curated whitelist; unknown IDs rejected
Facts	Fees and hours written by the model	Read from the verified dataset, never from the model
Routing	Arbitrary ordering of stops	Geographic clustering, under 4 h driving a day, 08:30–19:30
Budget	A vague range, unexplained	Itemised: stay + food + transport + entry equals the total
Kannada	Machine-translated, often broken	Human-curated names and phrases, with native audio
If AI fails	An error, and no plan	A rule-based planner takes over — it runs with no API keys

THE KEY DIFFERENCE

The model only orders verified places; every fact comes from the dataset, and rules take over when AI can't answer.

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APPROACH

How Dari Builds a Plan



01

Curate

39 verified places and 15 Kannada phrases, sourced from ASI, UNESCO, Karnataka Tourism, OpenStreetMap and IMD.

02

Score

Rank every place by interest match, month fit, budget tier and distance from the starting city.

03

Constrain

The model sees only that shortlist, and may answer with place IDs, a start time and one short note.

04

Validate

Zod schema plus rules: no unknown or repeated places, under 4 h driving, every day inside 08:30–19:30.

05

Enrich

Haversine distance × 1.3 road factor at 40 km/h, itemised cost, and Kannada names and phrases attached.

06

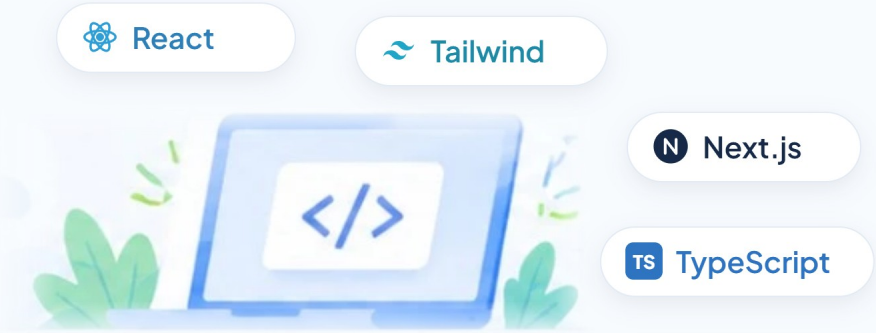
Fall back

Groq, then Gemini, then the rule-based planner, then a curated sample. It cannot return nothing.

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TECH STACK

What Dari Is Built On



FRAMEWORK
Next.js 16 App Router

LANGUAGE
TypeScript 5, strict

INTERFACE
React 19, Tailwind v4

MAPS
Leaflet, OpenStreetMap

MODELS
Groq Llama 3.3, Gemini

VALIDATION
Zod 4 on every boundary

SPEECH
Server proxy, Web Speech

SHARING
LZ-String, iCalendar

TESTING
Vitest, fully offline

DELIVERY
GitHub Actions, Vercel

39 curated places, every fact verified

128 automated tests across 18 suites

4 generation tiers, so it never fails

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ARCHITECTURE DIAGRAM

Architecture and Data Flow

From traveller input to a finished itinerary — ordered by AI, verified by rules.



BROWSER

1

Traveller input

- Interests
- Days
- Budget
- Pace
- Month
- Group size

SERVER · NEXT.JS ROUTE /API/PLAN

2

API gateway

Rate limit · 32 KB guard · Zod validation

3

Candidate scoring

Scores 39 curated places into a shortlist

4

Generation cascade

24 s budget · one retry per tier · shortlist-checked

- AI — Groq Llama 3.3
- AI — Google Gemini
- OFFLINE — Rule-based planner
- OFFLINE — Curated sample

5

Enrichment

Road legs and timings · itemised cost · Kannada names and phrases

BROWSER

6

Trip view

- Route map
- Day-by-day plan
- Cost breakdown
- Kannada audio
- Share link

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CONCLUSION

Conclusion and Next Steps



Where it stands today

- Live on Vercel, continuous integration green on every push
- Personalised multi-day plans built only from verified places
- Itemised budgets, route maps and native Kannada audio
- Works end to end with no AI keys configured at all

Business model

- Booking commissions on stays, cabs, buses and tickets
- Freemium: planning stays free, Pro adds offline maps and group trips
- White-label licensing for KSTDC, hotels and tour operators
- Labelled partner listings shown beside the plan, never inside it

Future scope

- Live KSRTC and rail timings for inter-city legs
- Offline progressive web app with cached map tiles
- Monsoon and temple-crowd signals feeding the route
- Hands-free voice changes for travellers on the road

LIVE DEPLOYMENT
dari-karnataka-travel-planner.vercel.app

SOURCE CODE
github.com/varshithvardhan2006-09/dari-karnataka-travel-planner